EX NO:1 Butterworth IIR Filters

1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

Objectives:

1. To determine the minimum filter order and cutoff frequency, required to satisfy the design constraints.

2. To design and analyze the magnitude response, phase response, characteristics of the Butterworth IIR low-pass filter for the given passband and stopband specifications.

3. To analyze the impulse response, step response and group delay characteristics of the Butterworth IIR low-pass filter for the given passband and stopband specifications.

4. To examine the pole-zero plot and verify the stability and performance of the designed low-pass filter.

Procedure:

1. Open MATLAB and create a new script file in the MATLAB Editor.

2. Define the filter specifications such as passband ripple (Rp), stopband attenuation (Rs), passband frequency (Fp), stopband frequency (Fsb), and sampling frequency (Fs).

3. Calculate the normalized passband and stopband frequencies with respect to the Nyquist frequency (Fs/2).

4. Use the buttord() function to determine the minimum order (N) and normalized cutoff frequency (Wn) required for the Butterworth low-pass filter.

5. Design the Digital IIR Butterworth Low-Pass Filter using the butter() function and obtain the numerator (b) and denominator (a) filter coefficients.

6. Display the filter order and cutoff frequency obtained from the design process.

7. Compute the frequency response of the designed filter using the freqz() function.

8. Plot the magnitude response of the filter and observe the passband characteristics and stopband attenuation.

9. Plot the phase response of the filter and analyze the phase variation over the frequency range.

10. Generate the pole-zero plot using the zplane() function to verify the stability of the designed IIR filter.

11. Create a test signal consisting of low-frequency and high-frequency components.

12. Apply the designed IIR low-pass filter to the test signal using the filter() function.

Program (Matlab Code):

clc; clear

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```
Fs=10000; Fp=1000; Fst=1500;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn);

freqz(b,a) % Magnitude & Phase

figure
[h,n]=impz(b,a,50);
...title('Impulse Response')

figure
stepz(b,a)

figure
zplane(b,a)

fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

Input:

- Passband Ripple (Rp) = 1 dB
- Stopband Attenuation (Rs

) = 20 dB

- Passband Frequency (Fp) = 1000 Hz

⍰ Stopband Frequency (Fs

) = 3000 Hz

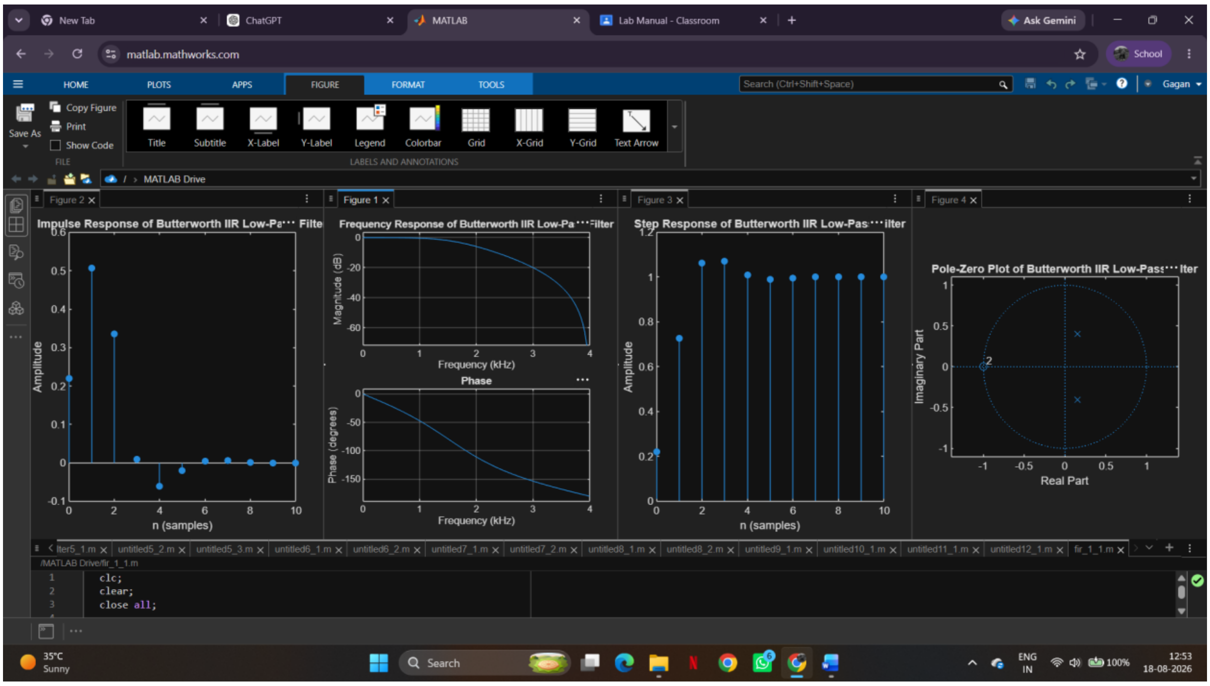
⍰ Sampling Frequency (Fsampling) = 8000 Hz

⍰ Filter Choice = Low-Pass Filter (LPF)

Output:

Minimum Filter Order (N) = 17

Cutoff Frequency (Wn) = 0.2083



Result:

1.The minimum filter order and cutoff frequency satisfying the given specifications were determined successfully.

2.The magnitude and phase responses of the Butterworth IIR low-pass filter were obtained and analyzed.

3.The impulse response, step response, and group delay characteristics were evaluated successfully.

4. The pole-zero plot confirmed the stability and proper performance of the designed filter.